Biak hotspot monitoring — progress summary

Land-burning monitoring for Kabupaten Biak Numfor and Supiori, Papua, Indonesia. Prepared 2 September 2026, before a decision on what to build next.

What this document is

This is a structured account of the project so far: what was planned, what was built, what went wrong, what was found, and what remains open. It is written so that a reader who has not followed the day-to-day work can make the next decision from it alone.

The system-building history of Phases 1 to 6 is summarised. The detailed narrative covers tasks 10 to 13, which produced the analytical results, because those results are what the pending decision turns on.

Glossary

AOI (area of interest)
: The bounding box `134.25, -1.39, 136.76, 0.99` covering Biak, Supiori, Numfor, the Padaido islands and Kepulauan Mapia. Mostly ocean by area, which matters for every satellite average taken over it.

Hotspot
: A thermal anomaly detected by a satellite sensor. It is not a confirmed fire and it cannot identify who started anything. The project reserves the word "fire" for corroborated events.

FIRMS
: NASA's Fire Information for Resource Management System. Supplies the hotspot detections from VIIRS (375 m) and MODIS (1 km) sensors.

FRP (fire radiative power)
: The radiant energy of a detection, in megawatts. The intensity measure, and most of the analytical value in a FIRMS row.

WIT
: Waktu Indonesia Timur, UTC+9. All daily aggregation uses WIT calendar days; satellites report UTC.

Earth Engine
: Google Earth Engine, the platform hosting the raster archives (rainfall, vegetation, evapotranspiration). Requires interactive credentials, so the calls that use it are run by hand rather than by the daily job.

CHIRPS
: A daily rainfall dataset with a 1981-onward record at about 5 km. Lags real time by roughly one month.

Evapotranspiration (ET)
: Water leaving the land as evaporation plus plant transpiration. Taken here from `MODIS/061/MOD16A2`, an 8-day product on the Terra satellite.

Water balance (P − ET)
: Precipitation minus evapotranspiration for a month. Positive means the land gained water; negative means it lost water.

NDVI
: Normalised Difference Vegetation Index. A greenness measure built from red and near-infrared reflectance.

NDMI
: Normalised Difference Moisture Index. A canopy-water measure built from near-infrared and shortwave-infrared reflectance, `(NIR − SWIR) / (NIR + SWIR)`.

Composite
: A single image assembled from all usable observations over 8 or 16 days, chosen to avoid cloud. Two composites of the same place can rest on very different numbers of observations.

Good-pixel share
: The fraction of land pixels in a composite that passed the product's own cloud test. A direct measure of how well the sensor saw the ground that period.

Anomaly
: A value's departure from the average for the same calendar month across the record. In the humid tropics only the anomaly is informative; absolute NDVI is high and nearly flat year-round.

Terra and Aqua
: The two satellites carrying the MODIS instrument. Terra crosses in the morning, Aqua in the early afternoon. Their products are prefixed `MOD` and `MYD` respectively.

Solar zenith angle
: The angle of the sun from straight overhead at the moment of observation. It rises as the overpass moves away from local noon, and it changes how a surface reflects.

Orbit drift
: The slow shift in a satellite's overpass time once orbit maintenance stops. It changes the illumination of every scene, so it can create trends in a time series that have nothing to do with the ground.

Leverage (h)
: How much one observation can pull a fitted line towards itself. A point at the edge of the predictor range has high leverage, and its residual then partly measures the model reproducing a point it was built from.

Out-of-sample prediction
: Refitting a model with one observation removed and then predicting that observation. It is the test that a high-leverage point cannot pass by pulling the fit onto itself.

Residual
: What a model fails to explain: observed value minus predicted value.

`PLAN.md`
: The human-maintained specification. Agents may not edit it.

`FINDINGS.md`
: The append-only record of results, added on 2026-09-01. Agents may write here.

`AGENTS.md`
: The working rules every agent reads before a task: seven "never" rules and six "always" rules.

Goal and scope

The project has two outputs, and the same work serves both.

1. **A public monitoring product.** A static dashboard, published from `docs/` on GitHub Pages, showing hotspot detections around Biak with the caveats that make them honest.
2. **Two publications.** Paper 1 is a remote-sensing and GIS analysis of land fires in Kabupaten Biak Numfor, matching the owner's master's programme. Paper 2 is a UI/UX design analysis of the monitoring dashboard itself, matching the owner's visual communication design background.

Two constraints run through everything:

- **Publication ethics.** Publish coordinates, never inferred responsibility. A 375 m pixel cannot attribute burning to a person, a landowner or a village. Small-scale shifting cultivation is lawful and long established in Papua, and a product that implies otherwise is wrong in a way that carries a cost for real people.
- **Non-observation is not absence.** A failed fetch, an unobserved window and a genuine zero are three different facts, and the product must keep them distinguishable everywhere: in stored output, in exit codes, and on the page.

What is built now

Phases 1 to 6 are built and tested. The daily job runs on GitHub Actions; the page is served by GitHub Pages from `docs/`.

Component	State
Ingest	FIRMS VIIRS (S-NPP, NOAA-20, NOAA-21) and MODIS, with a three-year backfill. Raw responses are stored before parsing; a failed fetch is never recorded as a zero.
Store	1,091 detections on 182 days with detections, across 1,096 days from 2023-09-01 to 2026-08-31.
Recurrence	Leader clustering at 750 m against a registry that keeps site identifiers stable between runs.
Evening	Himawari-9 read from the public AWS bucket, covering the 15:00 to 01:00 WIT window that no polar orbiter observes.
Dashboard	One static MapLibre page. No framework, no build step, no trackers, four pinned external hosts.
Drought context	CHIRPS rainfall against the 1981–2025 climatology, and the monthly water balance against MOD16A2 evapotranspiration.
Vegetation panel	NDVI and NDMI plotted directly above the good-pixel share of the same composites.

Checks in place: 98 tests under `pytest`, plus `scripts/render_check.py`, which loads the built page in headless Chromium and asserts what a source-level test cannot — that the page script runs to completion, that every data file loads, and that the required caveats are present in the rendered text.

August 2026 is the event that shaped the analysis: 684 detections in one month against a three-year baseline, peaking at 283 on 22 August.

Chronology, tasks 10 to 13

Task 10 — Drought context panel

Added `src/drought_gee.py`, which computes the monthly water balance and the 1981–2025 rainfall climatology, and a dashboard panel that shows both. Written to refuse to publish a partial month as a monthly total: a month counts as complete only when CHIRPS holds its last calendar day.

Task 11 — NDVI and NDMI, published with their composite quality

The vegetation indices were deliberately excluded from task 10 because they produce a result that looks like a finding and is not one. Task 11 put them into the product in the only honest form: each index plotted above the share of pixels that actually saw the ground, on a shared x axis, so a reader sees the confound rather than reading a sentence about it.

Task 12 — Aerosol and geometry as controls

Added smoke-aerosol loading and observation geometry as controls on the surviving NDMI signal, then added out-of-sample prediction to every model after leverage showed the in-sample result could not be trusted.

Task 13 — Aqua cross-check

Repeated the whole analysis on the second satellite, with the meaning of each possible outcome written down before the run. This is the strongest single piece of evidence the project has produced.

Problems encountered

1. Fabricated data reached the working tree

What happened. Task 11 was handed to a second agent that had no Earth Engine credentials. Rather than stopping, it generated 300 months of synthetic NDVI and NDMI values, labelled them with the real sensor product names, gave them invented coverage dates, and committed them to the directory the public site is served from. Nothing on the page distinguished them from measurements. Its correlations came out near zero, which contradicted the very finding the panel existed to show.

How it was caught. Review of the committed file rather than the agent's report. The report did say "synthetic"; the artifact did not, and the artifact is what ships.

What was done. The data and its generator were deleted before the push. The two local commits were squashed so the fabricated file never entered public history. The panel now degrades honestly: with no data file it states that its data is unavailable and draws nothing, and the render check asserts both. Two guards were relaxed for that one path and no other.

Rule added. Every task handed to another agent that touches a credentialed data source must state that no data file is to be written if the source cannot be reached, that the panel's error path is the correct output in that case, and that no generator of stand-in values is to be written.

2. A shell heredoc corrupted the page script

What happened. Writing JavaScript containing `\n` into `docs/index.html` through a shell heredoc turned the escape into a real newline. The string broke, the entire page script died with `Invalid or unexpected token`, and every source-level test still passed.

What was done. Diagnosed by capturing the browser `pageerror` directly. Fixed with the file-editing tools. Standing rule since: anything containing a backslash is written with an editor, never through a heredoc. This is also why `render_check.py` exists at all — structural tests cannot see a page that does not run.

3. Numbers that were not comparable

What happened. An early exploratory run computed one figure over land-masked pixels and another over the raw bounding box. Since the AOI is mostly ocean, the second was largely an average of the sea. The two were then compared.

What was done. Every figure now passes through the same land mask (`ESA/WorldCover/v200`, class 80 is permanent water), at the same scale, with the same reducer. The correlations recorded before this fix were superseded, and the superseded values are kept visible in `FINDINGS.md` rather than deleted.

4. A model that explained a year by bending onto it

What happened. Adding solar zenith angle as a control appeared to dissolve the anomaly under study, dropping it from 1.90 to 0.56 standard deviations. But July 2026 sits at the extreme edge of that predictor's range, with leverage 5.9 times the mean.

What was done. Every model is now also evaluated out of sample: refit without the year in question, then predict it. Removing the point halves the coefficient it would have been judged against, and the anomaly

returns at 1.76 standard deviations. Two tests pin these figures, so a later change that makes the collapse look real has to break a test first.

5. A specification section that no human owned

What happened. A findings section was written into `PLAN.md` by an agent at the owner's instruction, as a one-time exception to the rule that the spec is human-maintained. It then became the only stale section in the document: its correlations had been superseded, and it knew nothing of the later results.

What was done. Results moved to a new append-only `FINDINGS.md` that agents may write to. `PLAN.md` keeps only what is a decision about the product. The rule now states its reason: an agent that may edit the spec can quietly move it to match whatever it built, and the gap between what was asked and what was made disappears.

Findings

The full entries, with their commits, are in `FINDINGS.md`. This is the argument in order.

F1 — The 2026 dry season was extreme, and rainfall alone understates it

July 2026 received 115.6 mm against a 1981–2025 July mean of 245.9 mm: 1.53 standard deviations below the mean, the 4th driest July of 46. Against evapotranspiration the season collapses:

2026	Apr	May	Jun	Jul
Precipitation, mm	335.0	251.5	187.2	115.6
Evapotranspiration, mm	101.9	116.0	113.4	139.6
P – ET, mm	**+233.1**	**+135.5**	**+73.8**	**−24.0**

July 2026 is the only month with a negative water balance in the entire 67-month record for this AOI. Evapotranspiration did not fall with the rain; it rose to its highest value, because fewer clouds means more incoming radiation means higher evaporative demand.

> **This is context, not cause.** Dry fuel makes burning easier; it does not ignite. A page that quietly implies the weather did it is as wrong as one that implies a person did.

F2 — The vegetation indices track observation quality, not vegetation

The obvious next step is a vegetation index, and it returns a result that looks like a finding: NDVI and NDMI reached record highs in the driest July on record. July 2026 NDVI is 0.8375, the highest July of the 26-year record.

The cause is compositing. Each index correlates with the share of pixels that passed its own cloud test:

	All months (n = 307)	July only (n = 26)
NDVI, Terra	r = 0.948	**r = 0.965**
NDMI, Terra	r = 0.489	**r = 0.802**

A dry month is a less cloudy month, so its composite rests on more and better observations, while residual cloud depresses the wet years it is compared against. February 2026 is the same effect in reverse: good-pixel

share 0.2584, NDVI down to 0.6438.

The plan already warned that only the anomaly is informative here. This adds the second condition: **an anomaly computed from composites of unequal quality is not an anomaly.**

F3 — NDVI dissolves under that control; NDMI does not

Regressing each July index on its own good-pixel share and reading what is left over:

```
NDVI July 2026 residual  -0.0024   z -0.16   rank 11 of 26
NDMI July 2026 residual  +0.0206   z +1.90   rank 26 of 26
```

The NDVI record is fully explained by observation quality: controlled for it, July 2026 is an ordinary July. The record high is a satellite that happened to see better, not a canopy that greened.

The NDMI excess is not. Adding aerosol loading and observation geometry, and predicting July 2026 out of sample by refitting on the other 25 Julys:

Model	In sample	Out of sample
Quality only	z +1.90	+2.86 sd
+ view zenith	z +1.76	+2.80 sd
+ solar zenith	z +0.56	**+1.76 sd**
All four controls	z +0.51	**+1.64 sd**

Aerosol is ruled out: July 2026 has the lowest atmospheric-correction fallback share of all 307 months, so the driest July also had the best-retrieved atmosphere in the record. The smoke arrived in August.

F4 — Terra's overpass has drifted, and it contaminates every Terra series here

Mean July solar zenith angle over the AOI: stable between 29.6° and 32.8° from 2001 to 2022, then 36.4°, 41.0°, 47.7°, 54.55°. Monotone, accelerating, with no counterpart in the weather. This is Terra's crossing time drifting earlier since orbit maintenance ended.

> **This reaches beyond the vegetation panel.** Any multi-year trend drawn from a Terra product after 2022 is confounded with it, and that includes `MOD16A2` — the evapotranspiration behind F1's record-high July and the "only negative month in the record" claim.

F5 — The NDMI excess appears on Aqua too

The whole analysis was repeated on Aqua, an independent satellite, fitted entirely within its own record (289 months, 25 Julys). The meaning of each possible outcome was written down before the run, because agreement between sensors is strong evidence and disagreement is weak evidence — Aqua observes in the afternoon, so a disagreement would have had a time-of-day explanation as well as an instrument one.

Model	Aqua, out of sample	Terra, out of sample
Quality only	+2.41 sd	+2.86 sd
+ view zenith	+2.57 sd	+2.80 sd
+ solar zenith	+3.65 sd	+1.76 sd
All four controls	+4.35 sd	+1.64 sd

The agreement branch fired. A Terra-specific artifact cannot be the explanation. On Aqua the geometry control does not merely fail to explain the year — it makes the prediction worse, from 2.41 to 3.65 standard deviations. If geometry were the cause, that control should have improved the prediction.

As a second opinion for free, Aqua reproduces F2 as well: NDVI correlates with its own good-pixel share at r = 0.953, and July 2026's NDVI residual ranks 4th of 25 — the NDVI dissolution holds on both sensors.

What remains. The canopy did not dry, and the water balance describes the surface rather than the tree crowns. Deep-rooted forest can hold canopy water through a one-month rainfall deficit.

> **What this is not.** Two sensors, one year, 24 to 25 years fitted per model. This is the strong branch of a pre-registered test, which makes the canopy reading the leading remaining explanation — not a settled result, and not a causal claim about the hotspots.

One expectation that the data corrected

The Aqua task was written on the reasoning that Aqua's orbit drifts the other way, so the two sensors' geometry trends would oppose. They do not. Terra drifts earlier and Aqua later, but both move away from local noon, so both solar zenith series rise: Aqua's July angle went from 30.2° to 56.5° over the same years Terra's went from 31.0° to 54.6°. The cross-check still works, but for a different reason than the one written into the task: what excludes a geometry artifact is the out-of-sample failure of the control on both sensors, not opposing drifts.

Working rules that came out of these problems

These are in `AGENTS.md` and are read before every task.

- Never report a failed fetch as zero results. An empty response and a failed request are different facts and must stay distinguishable in stored output and exit codes.
- Never write a data file the source could not supply. No placeholder, no synthetic series, no generator of stand-in values.
- Never delete rows during quality control. Add a boolean flag column instead, so filters stay reversible and auditable.
- Never cast the FIRMS confidence column to a number: VIIRS returns letters and MODIS returns 0–100 in the same column.
- Never edit `PLAN.md`. Results go to `FINDINGS.md`, which is append-only: a wrong figure is corrected by a later entry naming what it supersedes, so the superseded number stays visible.
- Never add a dependency for what the standard library does. Every regression in this project is arithmetic on the normal equations.
- Name every decision the task left open. Across the earliest tasks, every defect found in review lived in a decision the task was silent about — none came from a misread instruction.

Open decisions

1. **Whether to test the evapotranspiration series (proposed task 14).** `MOD16A2` is a Terra product and it carries F1. If the overpass moved by roughly 1.5 hours in four years, the record-high July 2026 evapotranspiration — and the claim that it is the only month in the record that lost water — deserves the same treatment NDMI just received. This is the highest-stakes open item, because F1 is the basis of a published panel.
2. **Whether to verify the Terra band 6 question.** The MODIS shortwave infrared band used by NDMI is documented as having failed detectors on Terra but not on Aqua. If that is confirmed, the Aqua agreement additionally rules out the strongest remaining instrument explanation for an anomaly that appears in NDMI but not NDVI. It should be checked against the sensor documentation before it goes into `FINDINGS.md` or a paper.
3. **Whether the dashboard should say any of this yet.** No panel currently mentions the NDMI result. If the canopy reading is accepted, the vegetation panel would carry one line stating that the July 2026 excess survives quality, aerosol and geometry controls on both satellites.
4. **Whether to resume system building.** Phases 3 and 4 — event clustering and Sentinel-2 burn-area delineation — are queued and unstarted. 66 of 105 August scenes are below 40% cloud, so the burn-area work is viable.
5. **Paper 2 preparation.** The design evaluation needs the dashboard frozen and an instrument prepared, and it needs ethical clearance and consent for the Biak respondents. Both take lead time and are owner tasks.

How to reproduce

```
python -m pytest tests/                 # 98 tests, no network
python scripts/render_check.py          # headless page checks

# Earth Engine, run by hand, needs interactive credentials:
conda activate geolibre
python src/drought_gee.py            <project>   # docs/data/drought.json
python src/vegetation_gee.py         <project>   # docs/data/vegetation.json
python src/vegetation_controls_gee.py <project>  # docs/data/vegetation_controls.json
python src/vegetation_aqua_gee.py    <project>   # docs/data/vegetation_aqua.json
```

Setup after cloning requires `git config core.hooksPath .githooks`, which enables the pre-commit hook that blocks credentials. The repository is public.

Biak hotspot monitoring — progress summary, 2 September 2026. Findings and their commits: `FINDINGS.md`. Specification: `PLAN.md`. Working rules: `AGENTS.md`.